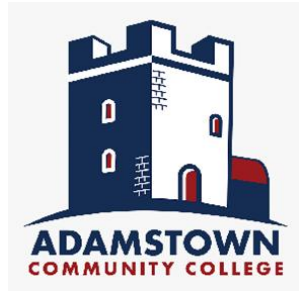


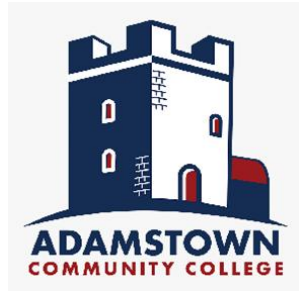
Lesson 06

Iteration



What is Iteration

- Iteration is the form of program control that allows us to instruct the computer to carry out a task over and over again by repeating a section of code, also called repetition, e.G. Taking in data from the user relating to multiple employees, students, repeated calculations etc.
- The program structure that is used to control this repetition is called a loop.
- There are two types of loops in python
 - While
 - For



The While Loop

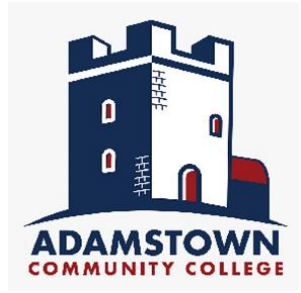
- The while statement continually executes a block of statements while a particular condition is true, similar structure to an if statement.
- The while statement evaluates the condition, which must return a boolean value, either true or false.

```
counter = 0  
while counter < 10:  
    print(counter)  
    counter += 1
```

Indentation is really important



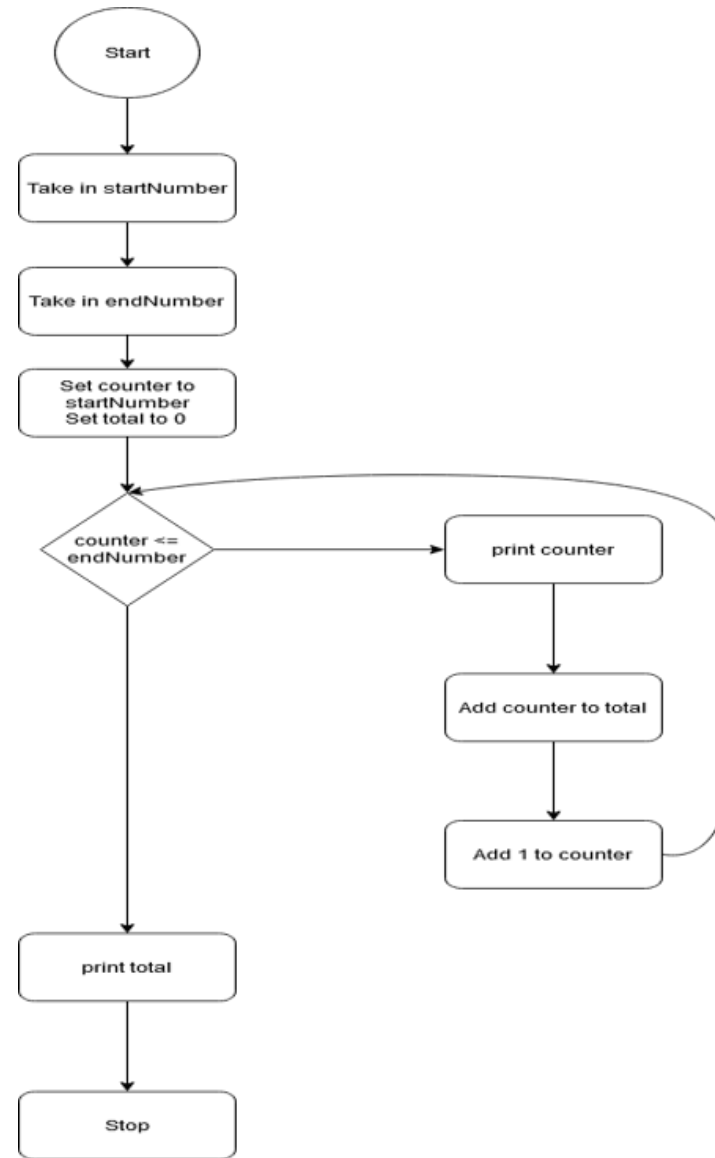
Indented code is run inside of the loop



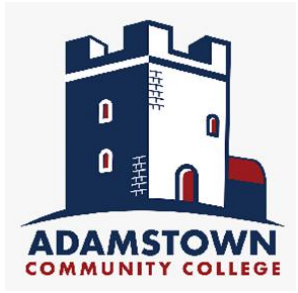
- Lets build a program which prints and counts from a start number to an end number.
- So, what inputs will our program need?
- Inputs: startNumber, endNumber
- So we want to print all the numbers between the startNumber and the endNumber and add all the numbers up too.

An algorithm is a set of steps to be followed to solve a problem.

This is what the algorithm for our program would look like



INDENTATION



```
WHILE COUNTER <= ENDNUMBER:  
    PRINT("THE COUNTER VALUE IS :", COUNTER)  
    TOTAL += COUNTER  
    COUNTER += 1
```

Notice the indentation! This is extremely important in python.

It indicates what block of code belongs to what statement.

```
STARTNUMBER = INT(INPUT("PLEASE ENTER A START VALUE: "))
ENDNUMBER = INT(INPUT("PLEASE ENTER AN END VALUE: "))
COUNTER = STARTNUMBER

TOTAL = 0

WHILE COUNTER <= ENDNUMBER:

    PRINT("THE COUNTER VALUE IS :",COUNTER)

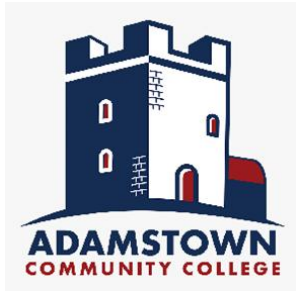
    TOTAL += COUNTER

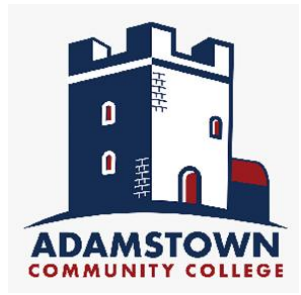
    COUNTER +=1

PRINT("TOTAL IS :",TOTAL)
```

- **INPUT:**

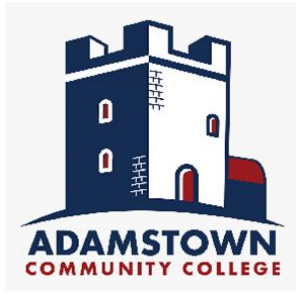
- STARTNUMBER = 1
- ENDNUMBER = 5





counter	total
1	1
2	3
3	6
4	10
5	15

```
The counter value is : 1
The counter value is : 2
The counter value is : 3
The counter value is : 4
The counter value is : 5
Total is : 15
```



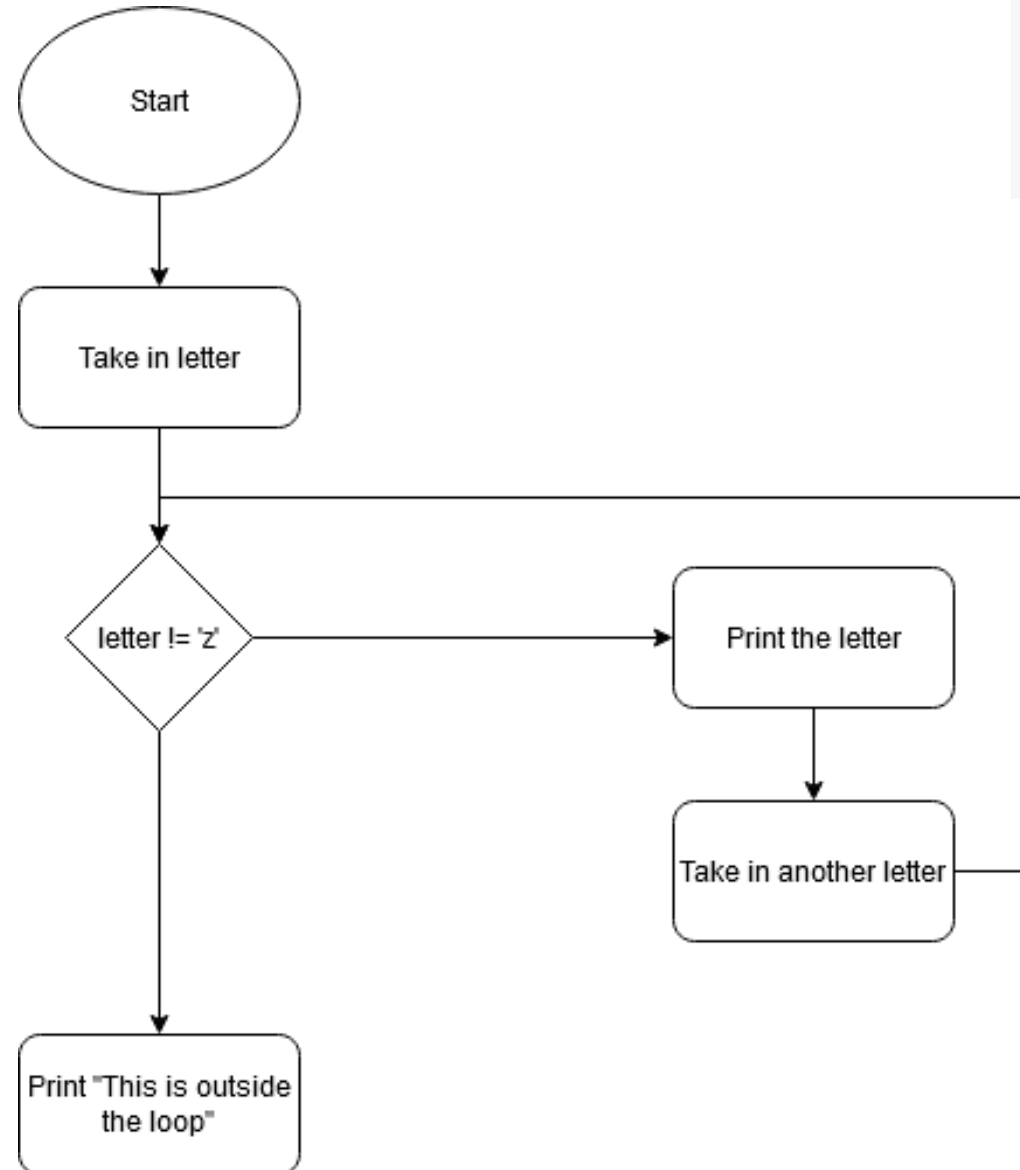
The While Loop

- Lets build another program which reads in characters from the keyboard. The program should stop when the letter 'z' is entered.
- So, what inputs will our program need?
- Inputs: inputLetter
- We want to end when the letter 'z' is entered.

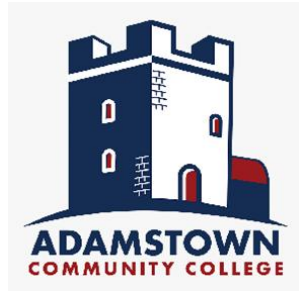
Algorithm

An algorithm is a set of steps to be followed to solve a problem.

This is what the algorithm for our program would look like



SO WHAT DOES THIS LOOK LIKE IN OUR CODE?



```
inputletter = input("please enter a letter: ")
while inputletter != 'z':
    print("the entered letter is :",inputletter)
    inputletter = input("please enter another letter: ")
Print("this is outside the loop")
```

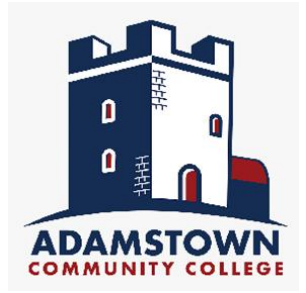
Please enter a letter: a

Please enter a letter: b

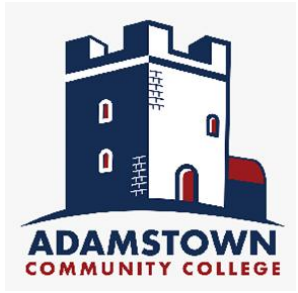
Please enter a letter: c

Please enter a letter: z

This is outside the loop



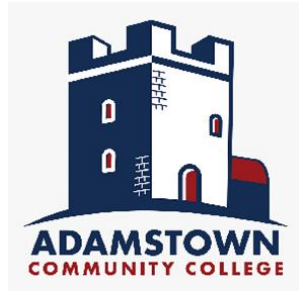
The For Loop



The For Loop

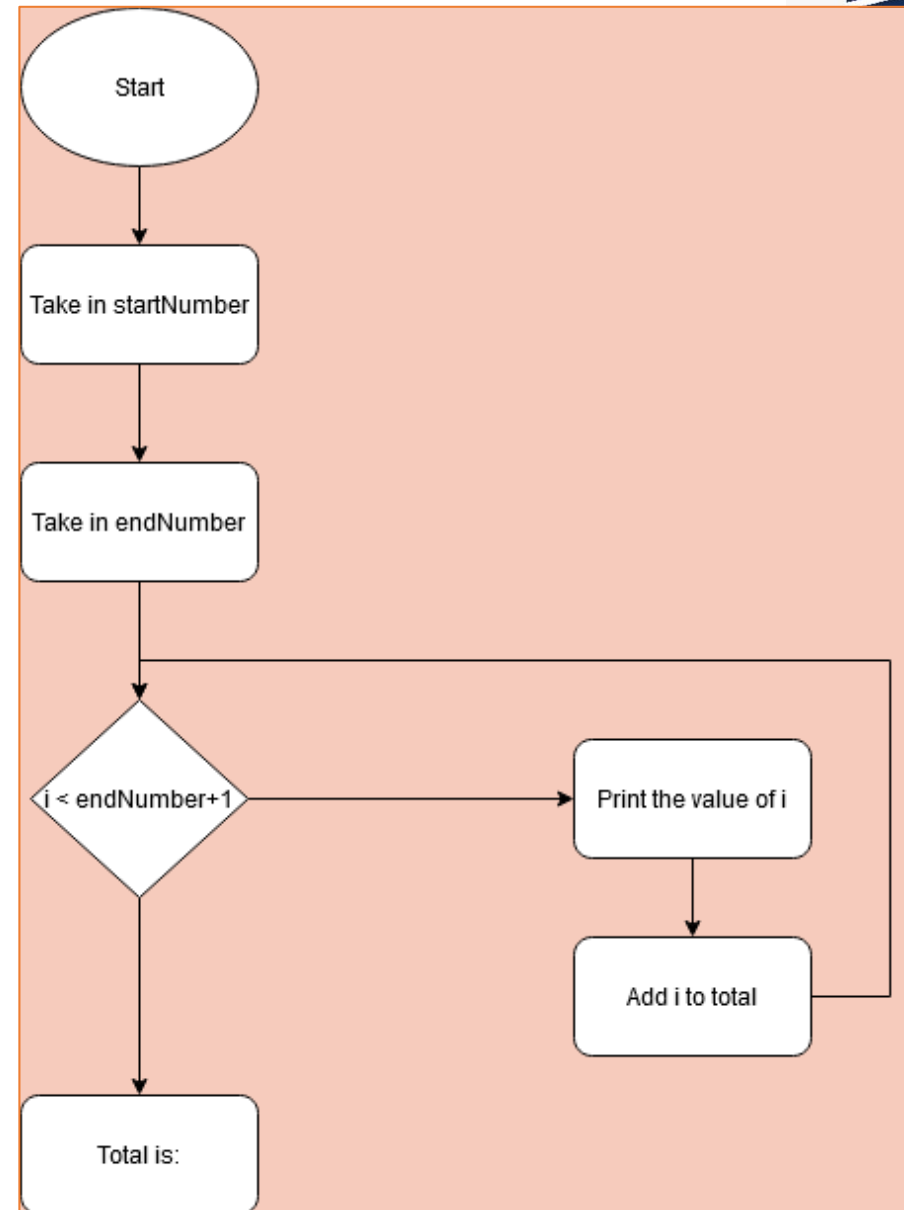
- Similar to the while the loop, the for loop will consist of the for keyword. The for loop works best for counting.
- With a for loop we use the range function to define a range to count between.

```
for i in range(1, 6):  
    print(i)
```



- Lets build a program which prints and counts from a start number to an end number.
- So, what inputs will our program need?
- Inputs: startNumber, endNumber
- So we want to print all the numbers between the startNumber and the endNumber and add all the numbers up too.

- An algorithm is a set of steps to be followed to solve a problem.
- This is what the algorithm for our program would look like




```
startNumber = int(input("Please enter a start value: "))
endNumber = int(input("Please enter an end value: "))
total = 0
```

```
for i in range(startNumber, endNumber+1):
    print("The value of i is:",i)
    total += i
print("Total is :",total)
```

Input:

Please enter a start value: 1

Please enter an end value: 10

Output:

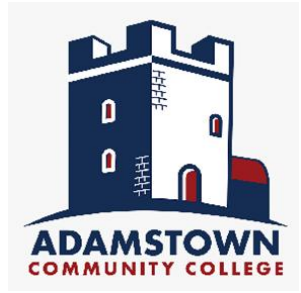
The value of i is: 1

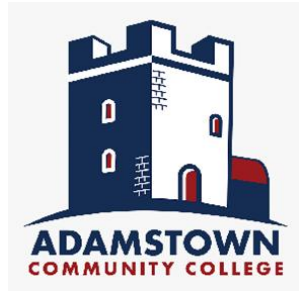
The value of i is: 2

..

The value of i is: 10

Total is : 55





i	total
1	1
2	3
3	6
4	10
5	15

The value of i is: 1

The value of i is : 2

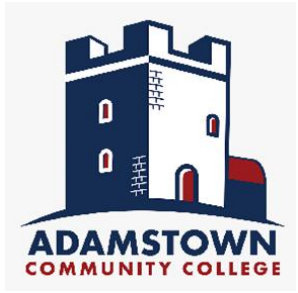
The value of i is : 3

The value of i is : 4

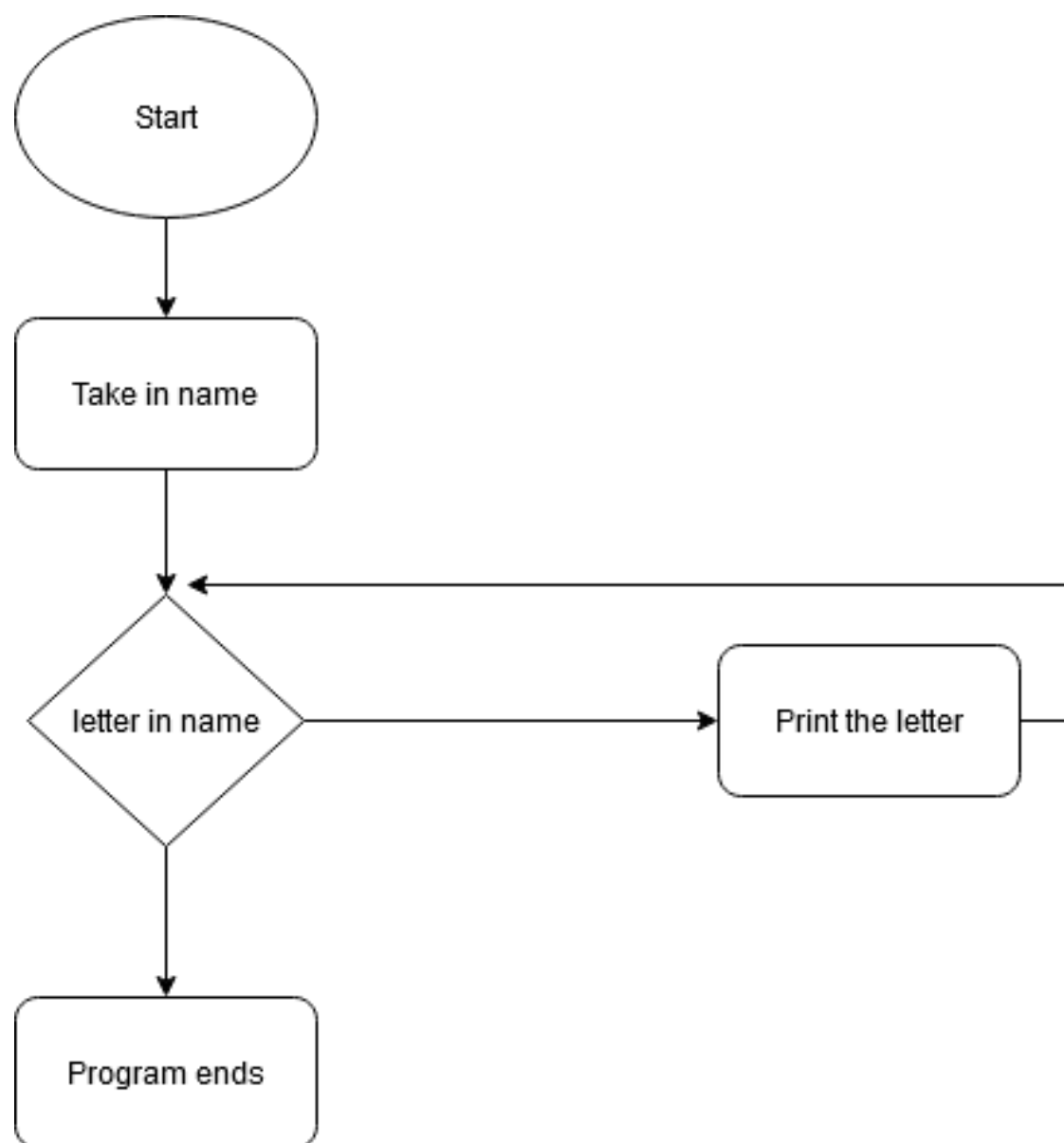
The value of i is : 5

Total is : 15

Another Loop



- Lets build a program which prints each character in a name
- So, what inputs will our program need?
- Inputs: inputName
- So we want to print all the characters in a string.



So what does this look like in our code?

```
inputname = input("please enter a name: ")
```

```
For letter i in inputname:
```

```
    print("the letter is:",letter)
```

Input:

```
Please enter a name: john
```

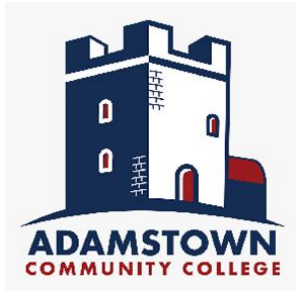
Output:

```
The letter is: j
```

```
The letter is: o
```

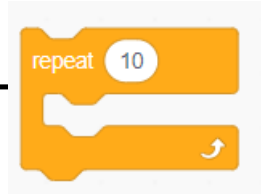
```
The letter is: h
```

```
The letter is: n
```

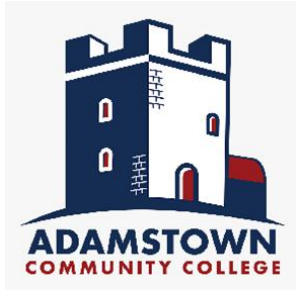
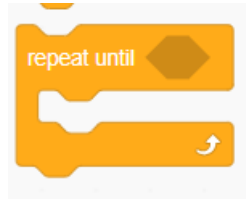


Scratch Equivalent

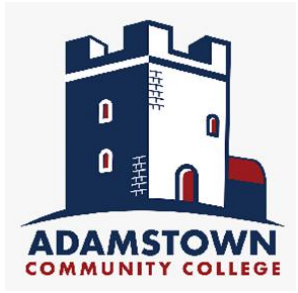
- For Loop -



- While -



Summary



Today you learned about:

- ✓ While loop.

 - ✓ Better suited for unknown endings like when 'z' is entered.

 - ✓ Loop ends when the condition is met.

- ✓ For loop.

 - ✓ Best for running between two particular numbers or a range or a set number of times.

 - ✓ Best for printing letters from a string.